

Reliable Cultural Knowledge Preservation in Multilingual LLMs through Model Merging

Hoang Quan Nguyen

WASP Media & Language, Umeå University, Sweden

QNGUYEN3@CS.UMU.SE

Nhut Huy Pham

VNG Corporation, Vietnam

HUYPN5@VNG.COM.VN

Maziyar Pahani

Arcee.ai, San Francisco, USA

MAZIYAR@ARCEE.AI

Johanna Björklund

Department of Computing Science, Umeå University, Sweden

JOHANNA@CS.UMU.SE

Xuan-Son Vu

WASP Media & Language, Umeå University, and DeepTensor AB, Sweden

SONVX@CS.UMU.SE

Abstract

We introduce a reliable approach for enhancing multilingual language models that preserves cultural knowledge while improving reasoning capabilities, focusing on low-resource languages. Using Qwen as a base model, we demonstrate that trust-aware model merging can verifiably improve performance without compromising cultural understanding. Our proposed approach achieves quantifiable improvements in both reasoning tasks and cultural benchmarks while maintaining computational efficiency. Results on Vietnamese and Arabic language tasks show consistent performance gains while preserving cultural knowledge, offering a reliable path for developing trustworthy multilingual AI systems. Our models are available at github.com/WARA-ML/waraml-mini-brains.

1. Introduction

The development of multilingual language models has primarily focused on high-resource languages, leaving significant gaps in performance for low-resource languages. While models like Qwen [Bai et al. \(2023\)](#) [Yang et al. \(2024\)](#) demonstrate strong multilingual capabilities, enhancing specific abilities like reasoning often comes at the cost of losing cultural knowledge. We propose a novel approach to this trade-off through strategic model merging.

We focus on two distinct languages: Vietnamese and Arabic. Vietnamese, with over 70 million internet users yet limited LLM training data, offers rich cultural nuances ideal for evaluating reasoning and cultural preservation. Arabic, with its morphological complexity and distinct script system, allows validation across linguistically diverse languages.

2. Related Work

2.1. Multilingual Language Models

Most non-English tasks face a significant challenge in accessing suitable language models, often having to choose between English-centric models with limited native language capabilities or closed-source solutions that restrict local deployment and customization. In the open-source domain, BLOOM and BLOOMZ ([Workshop et al. \(2023\)](#)) were early efforts supporting 46 languages, but showed performance gaps in complex reasoning tasks for low-resource languages. More recently, Qwen has emerged as a promising alternative, demonstrating strong performance in both reasoning and cultural understanding across multiple languages. However, the challenge of maintaining cultural knowledge while enhancing specific capabilities remains an open problem in the field [Pawar et al. \(2024\)](#).

2.2. Model Merging Techniques

Recent advances in model merging have provided efficient ways to combine multiple models while preserving their distinct capabilities. ModelSoup [Wortsman et al. \(2022\)](#) introduced the concept of weighted averaging of model parameters based on their task-specific performance. Linear merging [Matena and Raffel \(2022\)](#) demonstrated that simple weighted combinations of model weights can effectively transfer knowledge between models.

DARE [Yu et al. \(2024\)](#) advanced the field by introducing a consensus-based approach for merging task vectors, where parameters are combined based on their sign agreement across models, effectively filtering out conflicting updates. SLERP (Spherical Linear Interpolation) offered a geometrically motivated approach, interpolating between model weights along geodesics in the parameter space, which has proven particularly effective for preserving model capabilities during merging.

3. Methodology

We propose a general pipeline for merging LLMs while preserving cultural knowledge through supervised fine-tuning (SFT) and direct preference optimization (DPO). The method employs SLERP to interpolate between model parameters, enhancing one base model through SFT, performing an initial SLERP merge, then applying DPO training followed by a final SLERP operation. This approach enables effective knowledge transfer while maintaining parameter efficiency. Detail of the method is present in Algorithm 1.

3.1. Base Model Selection

We select Qwen-2.5-3B as a foundation model due to its strong multilingual capabilities, particularly its comprehensive understanding of Vietnamese language and culture. Unlike other open-source LLMs that primarily focus on English, Qwen demonstrates robust performance in low-resource languages without requiring additional language-specific pre-training.

3.2. Language-Specific Considerations

Our evaluation focuses on two linguistically diverse languages: Vietnamese and Arabic. For Vietnamese, we utilize the m-ArenaHard [CohereForAI \(2024\)](#) and VMLU [ZaloAI and JAIST](#)

Algorithm 1 Reliable Model Merging Pipeline with SLERP

Require:

```

1:  $\mathcal{M}_1, \mathcal{M}_2 \leftarrow$  Base models
2:  $\mathcal{D} \leftarrow$  Training samples
3:  $\alpha \leftarrow$  SLERP interpolation coefficient
4:  $\eta_{\text{sft}}, \eta_{\text{dpo}} \leftarrow$  Learning rates for SFT and DPO
5:  $N_{\text{sft}}, N_{\text{dpo}} \leftarrow$  Number of epochs
6: procedure PREPAREDATA( $\mathcal{D}$ )
7:    $\mathcal{D}_{\text{evolved}} \leftarrow$  Evolution( $\mathcal{D}$ )  $\triangleright$  Apply evolution strategy
8:    $\mathcal{D}_{\text{processed}} \leftarrow$  GenerateResponses( $\mathcal{D}_{\text{evolved}}$ ) return  $\mathcal{D}_{\text{processed}}$ 
9: procedure SLERP( $\mathcal{M}_a, \mathcal{M}_b, \alpha$ )
10:   $\theta_a \leftarrow$  Parameters( $\mathcal{M}_a$ )
11:   $\theta_b \leftarrow$  Parameters( $\mathcal{M}_b$ )
12:   $\omega \leftarrow \arccos(\langle \theta_a, \theta_b \rangle)$   $\triangleright$  Compute angular distance
13:   $\theta_{\text{merged}} \leftarrow \frac{\sin((1-\alpha)\omega)}{\sin(\omega)}\theta_a + \frac{\sin(\alpha\omega)}{\sin(\omega)}\theta_b$  return Model( $\theta_{\text{merged}}$ )
14: procedure MODELMERGING( $\mathcal{M}_1, \mathcal{M}_2, \mathcal{D}$ )
15:   $\mathcal{D}_{\text{train}} \leftarrow$  PrepareData( $\mathcal{D}$ )
16:   $\mathcal{M}_{\text{sft}} \leftarrow$  SFT( $\mathcal{M}_1, \mathcal{D}_{\text{train}}, \eta_{\text{sft}}, N_{\text{sft}}$ )
17:   $\mathcal{M}_{\text{merged}} \leftarrow$  SLERP( $\mathcal{M}_{\text{sft}}, \mathcal{M}_2, \alpha$ )
18:   $\mathcal{M}_{\text{dpo}} \leftarrow$  DPO( $\mathcal{M}_{\text{merged}}, \mathcal{D}_{\text{train}}, \eta_{\text{dpo}}, N_{\text{dpo}}$ )
19:   $\mathcal{M}_{\text{final}} \leftarrow$  SLERP( $\mathcal{M}_{\text{dpo}}, \mathcal{M}_2, \alpha$ ) return  $\mathcal{M}_{\text{final}}$ 

```

Ensure:

```

20:  $\mathcal{M}_{\text{final}}$  preserves cultural knowledge of both input models

```

(2023) benchmarks to test our models on both hard questions that require reasoning and cultural local understanding. In Arabic, we employ the OpenLLM Leaderboard benchmarks including ACVA, Alghafa, and EXAMS, which are specifically designed to evaluate Arabic language capabilities. This dual-language approach allows us to validate our methodology across distinctly different language families, scripts, and cultural contexts.

3.3. Training Pipeline

Our approach involves a multi-stage fine-tuning and merging process, illustrated in Algorithm 1. The process begins with dataset preparation, where we curate 20,000 mathematics and coding samples through an evolutionary complexity enhancement procedure, with responses generated using Claude 3.5 Sonnet. Regarding hyper-parameters, for both stages, we employed the AdamW optimizer with a weight decay of 0.1 and trained for 4 epochs using a batch size of 128. The SFT stage was performed on 20K samples with a learning rate of 2e-5, while the subsequent DPO stage utilized a larger dataset of 40K samples with a reduced learning rate of 2e-6.

3.4. Model Merging Strategy

We conducted SLERP for model merging using Mergekit (github.com/arcee-ai/mergekit) (Goddard et al. (2024)). For model parameters θ_a, θ_b and interpolation parameter ω , SLERP performs normalized angular interpolation (line 13 in Algorithm 1): $\theta_{\text{merged}} \leftarrow \frac{\sin((1-\alpha)\omega)}{\sin(\omega)}\theta_a + \frac{\sin(\alpha\omega)}{\sin(\omega)}\theta_b$, where $\omega = \arccos(\langle \theta_a, \theta_b \rangle)$. For numerical stability, we fall back to linear interpolation when vectors are nearly collinear (dot product > 0.9995).

4. Experiments

4.1. Experimental Setup

Our experimental pipeline consists of three main stages. We begin with Qwen-3B as our base model, chosen for its strong multilingual capabilities. This is followed by our reasoning-enhanced variants, developed through SFT and DPO on carefully curated instruction sets. The final model is produced through SLERP interpolation, merging the reasoning-enhanced model with the base model to preserve cultural knowledge while maintaining improved reasoning capabilities. All experiments are conducted on a single NVIDIA H100 GPU with gradient checkpointing enabled to optimize memory usage.

4.2. Benchmarks

4.2.1. ARENA-HARD BENCHMARK RESULTS

We evaluate Vietnamese reasoning capabilities using m-ArenaHard benchmark with Claude 3.5 Sonnet as judge.

Table 1: Head-to-Head performance on Vietnamese m-ArenaHard.

Model	Win (%)	Draw (%)	Lose (%)
PhoGPT-4B-Chat	95.4	3.6	1.0
Vistral-7B-chat	80.0	13.5	6.5
VinaLlama3.1-8B-Instruct	78.4	2.6	19.0
Llama3.1-8B-Instruct	61.8	25.5	12.7
Qwen2.5-7B-Instruct	57.1	13.3	29.6

Our model outperforms both open-source and proprietary models, winning 57-95% of comparisons despite competing models’ larger sizes.

4.2.2. VMLU BENCHMARK RESULTS

We evaluate cultural knowledge using the Vietnamese Multitask Language Understanding (VMLU) benchmark across training stages. Scores are reported in Table 2.

Table 2: VMLU Performance Comparison (%)

Model	STEM	Humanities	Social Science	Other	Total
Qwen-2.5-3B-Instruct	48.57	54.27	59.19	51.98	52.70
Ours (Final)	49.91	54.94	60.01	52.92	53.68

4.2.3. ARABIC OPENLLM LEADERBOARD

As reported in Table 3, our model’s performance on Arabic language tasks was evaluated using the OpenLLM Leaderboard benchmark suite, which includes various tasks designed to test both reasoning and language understanding capabilities.

Table 3: OpenLLM Leaderboard Performance Comparison

Model	Avg	ACVA	Alghafa	EXAMS	ARC-C	ARC-E	BOOLQ
Qwen2.5-7B-Instruct	0.53	0.43	0.59	0.51	0.60	0.60	0.78
Arcee Spark	0.53	0.51	0.60	0.48	0.56	0.63	0.78
Llama-3.1-8b	0.45	0.51	0.51	0.51	0.42	0.43	0.72
Aya-23-8B	0.47	0.47	0.58	0.43	0.44	0.50	0.70
Jais-13b	0.36	0.41	0.44	0.24	0.26	0.29	0.62
AceGPT-v2-8B	0.50	0.50	0.60	0.52	0.52	0.59	0.62
Ours	0.56	0.48	0.62	0.52	0.66	0.70	0.72

4.2.4. RESULTS AND ANALYSIS

Our model demonstrates strong performance across both target languages, achieving competitive results in both Vietnamese VMLU (53.68% overall) and Arabic benchmarks (average score 0.56 on OpenLLM Leaderboard). This consistent performance across linguistically diverse languages suggests the effectiveness of our approach in preserving language-specific capabilities while enhancing reasoning skills.

The effectiveness of our approach is evident across all benchmarks. In Vietnamese tasks, our model achieves the highest win rate (95.4%) in m-ArenaHard while showing consistent improvements in VMLU performance across all categories (STEM +1.34%, Social Science +0.82%, overall +0.98%), demonstrating that enhanced reasoning capabilities do not come at the cost of cultural knowledge. For Arabic, our model maintains the best average score (0.56) and leads in 5 out of 7 categories on the OpenLLM Leaderboard (including ARC-C at 0.66 and ARC-E at 0.70), notably outperforming larger models like Qwen2.5-7B (0.53) and Jais-13b (0.36). This consistent superior performance across linguistically diverse benchmarks validates our model merging strategy’s effectiveness in preserving language-specific capabilities while enhancing reasoning skills.

5. Discussion and Conclusion

We present an effective approach for enhancing reasoning capabilities while preserving cultural knowledge through strategic model merging, demonstrated across two linguistically diverse languages: Vietnamese and Arabic. Our model achieves competitive performance while maintaining strong cultural understanding in both language contexts. This methodology offers a practical solution for developing high-performing, locally-adapted language models in low-resource languages without extensive data collection requirements.

References

- Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, Binyuan Hui, Luo Ji, Mei Li, Junyang Lin, Runji Lin, Dayiheng Liu, Gao Liu, Chengqiang Lu, Keming Lu, Jianxin Ma, Rui Men, Xingzhang Ren, Xuancheng Ren, Chuanqi Tan, Sinan Tan, Jianhong Tu, Peng Wang, Shijie Wang, Wei Wang, Shengguang Wu, Benfeng Xu, Jin Xu, An Yang, Hao Yang, Jian Yang, Shusheng Yang, Yang Yao, Bowen Yu, Hongyi Yuan, Zheng Yuan, Jianwei Zhang, Xingxuan Zhang,

- 131 Yichang Zhang, Zhenru Zhang, Chang Zhou, Jingren Zhou, Xiaohuan Zhou, and Tian-
132 hang Zhu. Qwen technical report, 2023. URL <https://arxiv.org/abs/2309.16609>.
- 133 CohereForAI. m-ArenaHard: A multilingual LLM evaluation dataset. [https://](https://huggingface.co/datasets/CohereForAI/m-ArenaHard)
134 huggingface.co/datasets/CohereForAI/m-ArenaHard, 2024. Hugging Face Datasets.
- 135 Charles Goddard, Shamane Siriwardhana, Malikeh Ehghaghi, Luke Meyers, Vlad
136 Karpukhin, Brian Benedict, Mark McQuade, and Jacob Solawetz. Arcee’s mergekit: A
137 toolkit for merging large language models, 2024. URL [https://arxiv.org/abs/2403.](https://arxiv.org/abs/2403.13257)
138 [13257](https://arxiv.org/abs/2403.13257).
- 139 Michael Matena and Colin Raffel. Merging models with fisher-weighted averaging, 2022.
140 URL <https://arxiv.org/abs/2111.09832>.
- 141 Siddhesh Pawar, Junyeong Park, Jiho Jin, Arnav Arora, Junho Myung, Srishti Yadav,
142 Faiz Ghifari Haznitrana, Inha Song, Alice Oh, and Isabelle Augenstein. Survey of
143 cultural awareness in language models: Text and beyond, 2024. URL [https://arxiv.](https://arxiv.org/abs/2411.00860)
144 [org/abs/2411.00860](https://arxiv.org/abs/2411.00860).
- 145 BigScience Workshop, :, Teven Le Scao, Angela Fan, Christopher Akiki, Ellie Pavlick,
146 Suzana Ilić, Daniel Hesslow, Roman Castagné, Alexandra Sasha Luccioni, François
147 Yvon, Matthias Gallé, Jonathan Tow, Alexander M. Rush, Stella Biderman, Albert
148 Webson, Pawan Sasanka Ammanamanchi, Thomas Wang, Benoît Sagot, Niklas Muen-
149 nighoff, Albert Villanova del Moral, Olatunji Ruwase, Rachel Bawden, Stas Bekman,
150 Angelina McMillan-Major, Iz Beltagy, Huu Nguyen, Lucile Saulnier, Samson Tan, Pe-
151 dro Ortiz Suarez, Victor Sanh, Hugo Laurençon, Yacine Jernite, Julien Launay, Mar-
152 garet Mitchell, Colin Raffel, Aaron Gokaslan, Adi Simhi, Aitor Soroa, Alham Fikri Aji,
153 Amit Alfassy, Anna Rogers, Ariel Kreisberg Nitzav, Canwen Xu, Chenghao Mou, Chris
154 Emezue, Christopher Kamm, Colin Leong, Daniel van Strien, David Ifeoluwa Adelani,
155 Dragomir Radev, Eduardo González Ponferrada, Efrat Levkovizh, Ethan Kim, Eyal Bar
156 Natan, Francesco De Toni, Gérard Dupont, Germán Kruszewski, Giada Pistilli, Hady
157 Elshahar, Hamza Benyamina, Hieu Tran, Ian Yu, Idris Abdulmumin, Isaac Johnson, Itziar
158 Gonzalez-Dios, Javier de la Rosa, Jenny Chim, Jesse Dodge, Jian Zhu, Jonathan Chang,
159 Jörg Froberg, Joseph Tobing, Joydeep Bhattacharjee, Khalid Almubarak, Kimbo Chen,
160 Kyle Lo, Leandro Von Werra, Leon Weber, Long Phan, Loubna Ben allal, Ludovic Tan-
161 guy, Manan Dey, Manuel Romero Muñoz, Maraim Masoud, María Grandury, Mario
162 Šaško, Max Huang, Maximin Coavoux, Mayank Singh, Mike Tian-Jian Jiang, Minh Chien
163 Vu, Mohammad A. Jauhar, Mustafa Ghaleb, Nishant Subramani, Nora Kassner, Nuru-
164 laqilla Khamis, Olivier Nguyen, Omar Espejel, Ona de Gibert, Paulo Villegas, Peter
165 Henderson, Pierre Colombo, Priscilla Amuok, Quentin Lhoest, Rheza Harliman, Rishi
166 Bommasani, Roberto Luis López, Rui Ribeiro, Salomey Osei, Sampo Pyysalo, Sebastian
167 Nagel, Shamik Bose, Shamsuddeen Hassan Muhammad, Shanya Sharma, Shayne Long-
168 pre, Somaieh Nikpoor, Stanislav Silberberg, Suhas Pai, Sydney Zink, Tiago Timponi
169 Torrent, Timo Schick, Tristan Thrush, Valentin Danchev, Vassilina Nikoulina, Veronika
170 Laippala, Violette Lepercq, Vrinda Prabhu, Zaid Alyafeai, Zeerak Talat, Arun Raja, Ben-
171 jamin Heinzerling, Chenglei Si, Davut Emre Taşar, Elizabeth Salesky, Sabrina J. Mielke,
172 Wilson Y. Lee, Abheesht Sharma, Andrea Santilli, Antoine Chaffin, Arnaud Stiegler,

173 Debajyoti Datta, Eliza Szczechla, Gunjan Chhablani, Han Wang, Harshit Pandey, Hen-
 174 drik Strobelt, Jason Alan Fries, Jos Rozen, Leo Gao, Lintang Sutawika, M Saiful Bari,
 175 Maged S. Al-shaibani, Matteo Manica, Nihal Nayak, Ryan Teehan, Samuel Albanie,
 176 Sheng Shen, Srulik Ben-David, Stephen H. Bach, Taewoon Kim, Tali Bers, Thibault
 177 Fevry, Trishala Neeraj, Urmish Thakker, Vikas Raunak, Xiangru Tang, Zheng-Xin Yong,
 178 Zhiqing Sun, Shaked Brody, Yallow Uri, Hadar Tojarieh, Adam Roberts, Hyung Won
 179 Chung, Jaesung Tae, Jason Phang, Ofir Press, Conglong Li, Deepak Narayanan, Hatim
 180 Bourfoune, Jared Casper, Jeff Rasley, Max Ryabinin, Mayank Mishra, Minjia Zhang, Mo-
 181 hammad Shoeybi, Myriam Peyrounette, Nicolas Patry, Nouamane Tazi, Omar Sanseviero,
 182 Patrick von Platen, Pierre Cornette, Pierre François Lavallée, Rémi Lacroix, Samyam Ra-
 183 jbandari, Sanchit Gandhi, Shaden Smith, Stéphane Requena, Suraj Patil, Tim Dettmers,
 184 Ahmed Baruwa, Amanpreet Singh, Anastasia Cheveleva, Anne-Laure Ligozat, Arjun Sub-
 185 ramonian, Aurélie Névél, Charles Lovering, Dan Garrette, Deepak Tunuguntla, Ehud
 186 Reiter, Ekaterina Taktasheva, Ekaterina Voloshina, Eli Bogdanov, Genta Indra Winata,
 187 Hailey Schoelkopf, Jan-Christoph Kalo, Jekaterina Novikova, Jessica Zosa Forde, Jor-
 188 dan Clive, Jungo Kasai, Ken Kawamura, Liam Hazan, Marine Carpuat, Miruna Clin-
 189 ciu, Najoung Kim, Newton Cheng, Oleg Serikov, Omer Antverg, Oskar van der Wal,
 190 Rui Zhang, Ruochen Zhang, Sebastian Gehrmann, Shachar Mirkin, Shani Pais, Tatiana
 191 Shavrina, Thomas Scialom, Tian Yun, Tomasz Limisiewicz, Verena Rieser, Vitaly Pro-
 192 tasov, Vladislav Mikhailov, Yada Pruksachatkun, Yonatan Belinkov, Zachary Bamberger,
 193 Zdeněk Kasner, Alice Rueda, Amanda Pestana, Amir Feizpour, Ammar Khan, Amy
 194 Faranak, Ana Santos, Anthony Hevia, Antigona Unldreaj, Arash Aghagol, Arezoo Ab-
 195 dollahi, Aycha Tammour, Azadeh HajiHosseini, Bahareh Behroozi, Benjamin Ajibade,
 196 Bharat Saxena, Carlos Muñoz Ferrandis, Daniel McDuff, Danish Contractor, David Lan-
 197 sky, Davis David, Douwe Kiela, Duong A. Nguyen, Edward Tan, Emi Baylor, Ezinwanne
 198 Ozoani, Fatima Mirza, Frankline Ononiwu, Habib Rezanejad, Hessie Jones, Indrani Bhat-
 199 tacharya, Irene Solaiman, Irina Sedenko, Isar Nejadgholi, Jesse Passmore, Josh Seltzer,
 200 Julio Bonis Sanz, Livia Dutra, Mairon Samagaio, Maraim Elbadri, Margot Mieskes,
 201 Marissa Gerchick, Martha Akinlolu, Michael McKenna, Mike Qiu, Muhammed Ghauri,
 202 Mykola Burynek, Nafis Abrar, Nazneen Rajani, Nour Elkott, Nour Fahmy, Olanrewaju
 203 Samuel, Ran An, Rasmus Kromann, Ryan Hao, Samira Alizadeh, Sarmad Shubber, Silas
 204 Wang, Sourav Roy, Sylvain Viguier, Thanh Le, Tobi Oyebade, Trieu Le, Yoyo Yang,
 205 Zach Nguyen, Abhinav Ramesh Kashyap, Alfredo Palasciano, Alison Callahan, Anima
 206 Shukla, Antonio Miranda-Escalada, Ayush Singh, Benjamin Beilharz, Bo Wang, Caio
 207 Brito, Chenxi Zhou, Chirag Jain, Chuxin Xu, Clémentine Fourrier, Daniel León Perinián,
 208 Daniel Molano, Dian Yu, Enrique Manjavacas, Fabio Barth, Florian Fuhrmann, Gabriel
 209 Altay, Giyaseddin Bayrak, Gully Burns, Helena U. Vrabec, Imane Bello, Ishani Dash,
 210 Jihyun Kang, John Giorgi, Jonas Golde, Jose David Posada, Karthik Rangasai Sivara-
 211 man, Lokesh Bulchandani, Lu Liu, Luisa Shinzato, Madeleine Hahn de Bykhovetz, Maiko
 212 Takeuchi, Marc Pàmies, Maria A Castillo, Marianna Nezhurina, Mario Sängner, Matthias
 213 Samwald, Michael Cullan, Michael Weinberg, Michiel De Wolf, Mina Mihaljcic, Minna
 214 Liu, Moritz Freidank, Myungsun Kang, Natasha Seelam, Nathan Dahlberg, Nicholas Mi-
 215 chio Broad, Nikolaus Muellner, Pascale Fung, Patrick Haller, Ramya Chandrasekhar, Re-
 216 nata Eisenberg, Robert Martin, Rodrigo Canalli, Rosaline Su, Ruisi Su, Samuel Cahyaw-
 217 ijaya, Samuele Garda, Shlok S Deshmukh, Shubhanshu Mishra, Sid Kiblawi, Simon Ott,

- 218 Sinee Sang-aroonsiri, Srishti Kumar, Stefan Schweter, Sushil Bharati, Tanmay Laud,
 219 Théo Gigant, Tomoya Kainuma, Wojciech Kusa, Yanis Labrak, Yash Shailesh Bajaj,
 220 Yash Venkatraman, Yifan Xu, Yingxin Xu, Yu Xu, Zhe Tan, Zhongli Xie, Zifan Ye,
 221 Mathilde Bras, Younes Belkada, and Thomas Wolf. Bloom: A 176b-parameter open-
 222 access multilingual language model, 2023. URL <https://arxiv.org/abs/2211.05100>.
- 223 Mitchell Wortsman, Gabriel Ilharco, Samir Yitzhak Gadre, Rebecca Roelofs, Raphael
 224 Gontijo-Lopes, Ari S. Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Si-
 225 mon Kornblith, and Ludwig Schmidt. Model soups: averaging weights of multiple
 226 fine-tuned models improves accuracy without increasing inference time, 2022. URL
 227 <https://arxiv.org/abs/2203.05482>.
- 228 An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li,
 229 Chengyuan Li, Dayiheng Liu, Fei Huang, Guanting Dong, Haoran Wei, Huan Lin, Jialong
 230 Tang, Jialin Wang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Ma, Jianxin Yang,
 231 Jin Xu, Jingren Zhou, Jinze Bai, Jinzheng He, Junyang Lin, Kai Dang, Keming Lu, Keqin
 232 Chen, Kexin Yang, Mei Li, Mingfeng Xue, Na Ni, Pei Zhang, Peng Wang, Ru Peng, Rui
 233 Men, Ruize Gao, Runji Lin, Shijie Wang, Shuai Bai, Sinan Tan, Tianhang Zhu, Tianhao
 234 Li, Tianyu Liu, Wenbin Ge, Xiaodong Deng, Xiaohuan Zhou, Xingzhang Ren, Xinyu
 235 Zhang, Xipin Wei, Xuancheng Ren, Xuejing Liu, Yang Fan, Yang Yao, Yichang Zhang,
 236 Yu Wan, Yunfei Chu, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, Zhifang Guo, and Zhihao
 237 Fan. Qwen2 technical report, 2024. URL <https://arxiv.org/abs/2407.10671>.
- 238 Le Yu, Bowen Yu, Haiyang Yu, Fei Huang, and Yongbin Li. Language models are super
 239 mario: Absorbing abilities from homologous models as a free lunch, 2024. URL <https://arxiv.org/abs/2311.03099>.
- 241 ZaloAI and JAIST. VMLU: A Vietnamese Multitask Language Understanding benchmark
 242 suite for large language models. <https://vmlu.ai/>, 2023. Accessed: 2024-03-10.